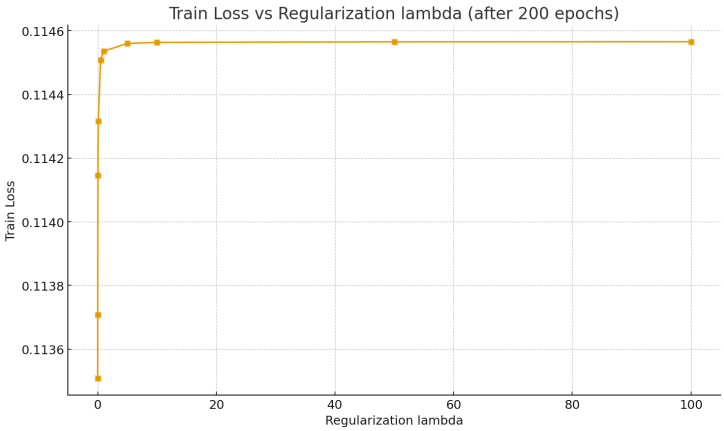
CPHC100 Project 1 Group Writeup

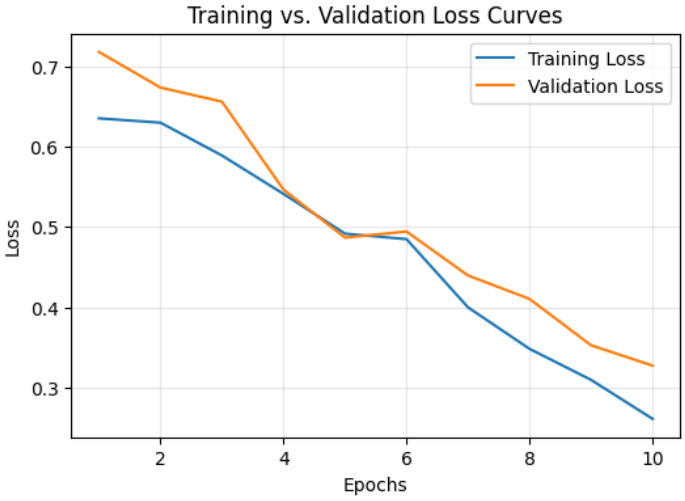
Jake Pastoria, Asli Alpman, Majed AlFaraj, Viraj Gupta

Q1.2 Regularization & Training Loss



This plot shows the behaviour of training loss (after 200 epochs) as a function of the regularisation lambda. As lambda increases, the model complexity decreases due to shrinking the weights toward zero. This reduces overfitting, hence increasing training loss. Once regularisation lambda becomes large enough, the weights get very close to zero. At this point, the training loss plateaus.

Q2.1 Ablation Study

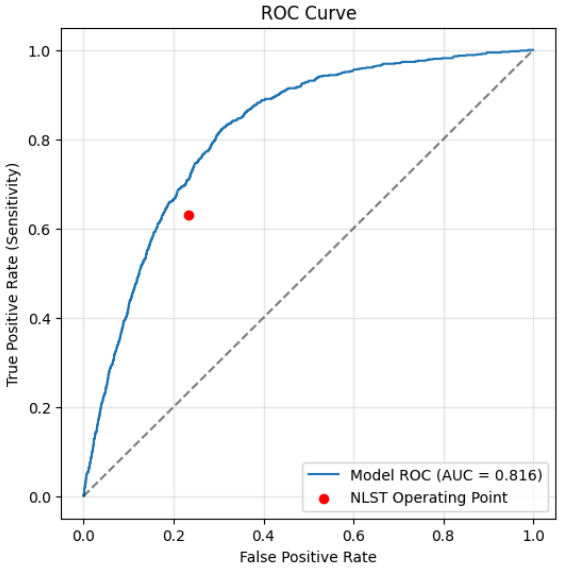


Training and validation losses both decreased and converged, without strong evidence of overfitting. Ten epochs were sufficient for convergence with our learning rate (0.01) and batch size (32) choices. This indicates that the model learns meaningful patterns from the data and does not overfit during training. The loss convergence means that the number of epochs (10) and batch size was an appropriate choice. Important design decisions for achieving convergence and optimal performance included:

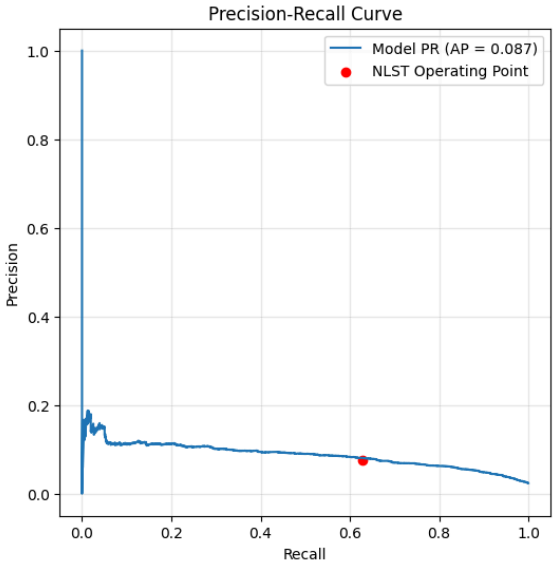
- Careful feature engineering with both numerical (e.g., age, pack_years), categorical (e.g., sex, race7, smoked_f), and ordinal features (e.g., educat, agelevel), transformed using a vectorizer for normalization and encoding.
- Using regularization for the model (lambda = 0.001).

- Leveraging stochastic gradient descent with well-chosen hyperparameters for proper learning on a large data split (100,000 train, 25,000 validation, 29,887 test).

Q2.2 Analyzing Model Performance



The ROC curve of the model shows an AUC of 0.816, showing that the model has a good ability to distinguish between those who will and will not develop lung cancer. The steep initial rise and curve above the diagonal means that the model is much better than random classification. The red dot marks the operating point of the current NLST criteria. The NLST operating point achieves a sensitivity (True Positive Rate) of approximately 0.63 while keeping the False Positive Rate near 0.2, indicating significant detection with the cost of false alarms.



The precision-recall curve, however, shows a challenge due to dataset imbalance. The average precision (AP) is 0.087. At the NLST operating point (recall ≈ 0.63), precision drops to roughly 0.10, meaning that only 10% of flagged cases are true positives. While recall is moderate (true lung cancers are detected), the trade-off is low precision, showing that many flagged as high-risk do not develop lung cancer. Both curves highlight that:

- The model has a robust AUC and controlled recall at chosen operating points.
- The current NLST criteria balance sensitivity and specificity but struggle with precision.

- Logistic regression trained via SGD, vectorized features, and grid search for hyperparameters yielded strong generalization and robust validation/test scores.
- Feature configuration such as normalization and one-hot encoding is crucial for maintaining stability in training.
- The NLST operating point analysis demonstrates the tradeoffs between improved detection and the rate of false alarms, crucial for actionable screening protocols.

Subgroup Analysis & Analysis Question:

Here are the results for each subgroup AUC, as well as an analysis on whether or not the difference was greater than 0.05:

Sex:

- Male: 0.796
- Female: 0.834

The AUC difference between the 'Male' and 'Female' respondents is around 0.038, so there doesn't seem to be a significant difference between sexes. There seems to be a nearly even split of women and men in the dataset, yet there is not an even split amongst men and women for those with lung cancer. There seems to be around 1.5x the number of men with lung cancer compared to women, which could affect the AUC as the model is likely biased towards predicting "no" cancer due to there being significantly more individuals without cancer.

Race:

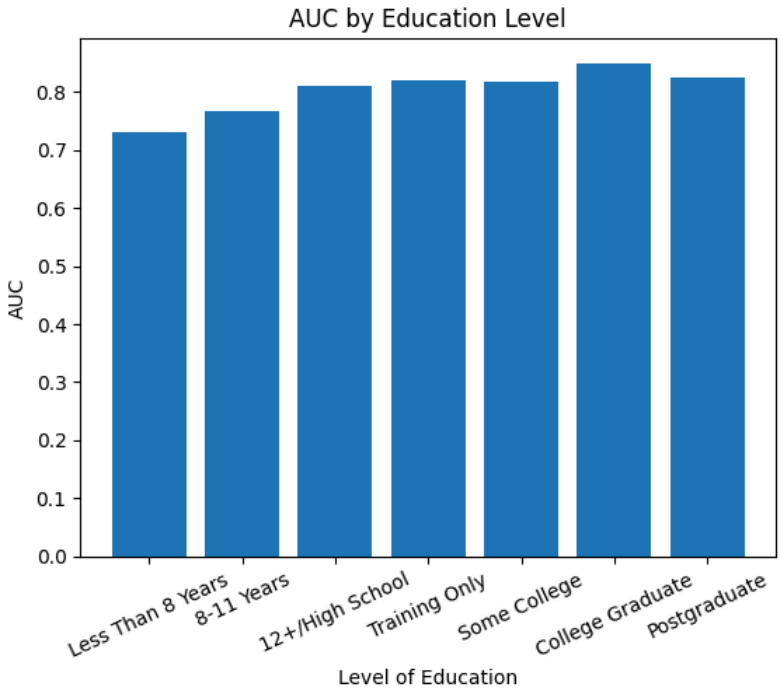
- White, Non-Hispanic: 0.827
- Black, Non-Hispanic: 0.79
- Hispanic: 0.749
- Asian: 0.807
- Pacific Islander: 0.889
- American Indian: 0.542
- Missing: 0.551

Race has a wide range of differences in AUC, with some being significantly greater than 0.05. This is extremely likely to be due to the smaller number of participants in some groups, most notably the "American Indian" and "Pacific Islander" categories, which make up ~0.2% and ~0.617% of the datasets, respectively. We can see that the AUC average is

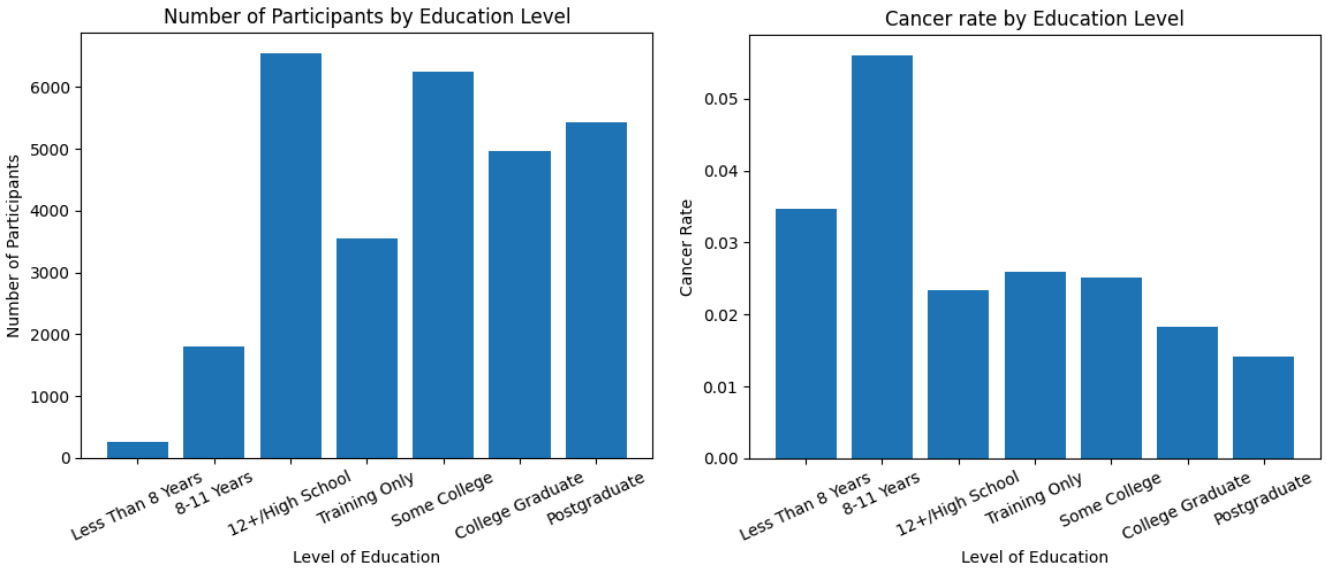
much higher for “White” and “Black” participants, while the average for these less common participants has a much higher variance.

Education:

- Less Than 8 Years: 0.731
- 8-11 Years: 0.765
- 12 Years Or Completed High School: 0.809
- Post High School Training Other Than College: 0.819
- Some College: 0.816
- College Graduate: 0.849
- Postgraduate: 0.824



The educational level of the participants followed an interesting trend: AUC generally seems to increase as the educational level increases. This relationship generally mirrors the number of participants in each educational level, and many of the differences between groups were greater than 0.05. In addition, it seems to be negatively correlated with the cancer rate in each educational level. Take the test set for example, shown below:



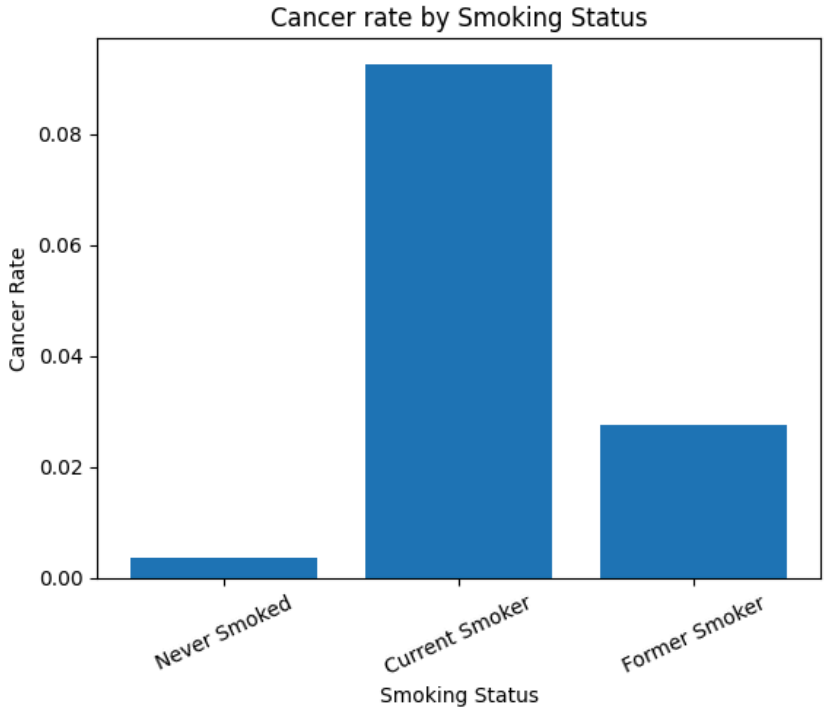
We can see that the number of participants in the validation set generally follows an upward trend (right figure), while the cancer rate by education level (left figure) seems to decrease as the participants' level of education increases.

Similar to the previous subgroups, the groups that have lower cancer rates in the data (such as the Postgraduates and College graduates) seem to have a higher AUC. This is once again likely due to the model's bias towards predicting "no cancer". Despite this, the level of variance seen in the first two categories due to the low number of participants makes it significantly difficult to draw any conclusions and limits our analyses.

Cigarette Smoking Status

- Never Smoked: 0.648
- Current Cigarette Smoker: 0.637
- Former Cigarette Smoker: 0.753

The difference in AUC shown here counters our previous notion that groups with lower cancer rates seem to have better AUCs in the test set. This graph here shows the lung cancer rates by smoking status:



Note: The "Current Smoker" category has far fewer participants than the other two categories, which could contribute to its high cancer rate/unique AUC.

We can see that the difference between the "Never Smoked" and "Former Smoker" groups has an AUC difference that is greater than 0.05. The high AUC for the smoker group likely means that an individual's smoking status is a good discriminator for cancer prediction; although it does seem that the model isn't as good at distinguishing non-smokers.

NLST Flag

- No: 0.782
- Yes: 0.64

Interestingly, the AUC between those who qualify and don't qualify for the NLST shows that the model performs much better on those who don't qualify. This result is also likely misleading, however, due to the large imbalance between the number of participants in the "Yes" subgroup. In the data, there are ~4x more individuals who don't qualify for the NLST, which means that most training will be done on those who don't qualify.
Overall, a majority of our analyses are limited by the overwhelming class bias towards those without cancer, as well as the lack of a diverse participant pool. This makes drawing conclusions and making an effective screening model difficult.

Q2.3 Model Interpretation

After analysing the weights of our chosen model, we determined that the most important features are:

1. “age” → (+0.6415)
2. “pack_years” → (+0.5255)
3. “smoked_f = 0” → (-0.6813) [OHE]

For the numerical columns “age”, “pack_years”, since the values were normalised, we can interpret the weight in the following way: if the age increases by +1 standard deviation, then the odds increase by a multiplicative factor of exp(0.6415) = 1.889 (1.69 for pack years). This strong positive correlation indicates that older individuals and smokers are more likely to have lung cancer.

Similarly, for “smoked_f”, we see a value of 0.6813. This strong negative association tells us that individuals who have never smoked before have significantly lower odds of developing lung cancer (exp(-0.6813) = 0.506 the odds).

Q2.4 Simulating Clinical Utility

Q2.4.1

To calculate sensitivity, specificity, and PPV, we define the values as follows:

- TP: lung_cancer=1 & nlst_flag=1
- FP: lung_cancer=0 & nlst_flag=1
- TN: lung_cancer=0 & nlst_flag=0
- FN: lung_cancer=1 & nlst_flag=0

After defining these values, then we can say that:

- Sensitivity = TP/(TP + FN) [recall]
- Specificity = TN/(TN + FP) [true negative rate]
- PPV = TP/(TP + FP) [precision]

We calculate the values as the following

- Sensitivity = 0.66666666

- Specificity = 0.8029895
- PPV = 0.07537012

Q2.4.2

To calculate the matched sensitivity performance when the NLST specificity is matched, we need to follow a few steps:

1. Get the TP, TN, FP, FN values for many different thresholds
    a. The threshold value is going to control the specificity and sensitivity values
2. Find the threshold value that gets our model's specificity closest to 0.8029895...
3. Find the corresponding sensitivity value using the threshold value found in the previous step

Doing so, we match the specificity (0.8029895...) at the threshold = 0.034259147, which corresponds to a **sensitivity = 0.694940476**.

Our model achieved a higher sensitivity when matching the specificity. This tells us that for the same false positive rate as NLST (by extension as FPR = 1 - TNR), our model correctly finds a bit more cancer (better recall).

Q2.4.3

We evaluated and compared several methods for selecting the optimal logistic regression threshold. The approaches we assessed are:

1. Default 0.5 Threshold: Using the standard 0.5 cutoff value

2. Maximum F1 Score: Selecting the threshold that yields the highest F1 score, which is the harmonic mean of precision and recall.

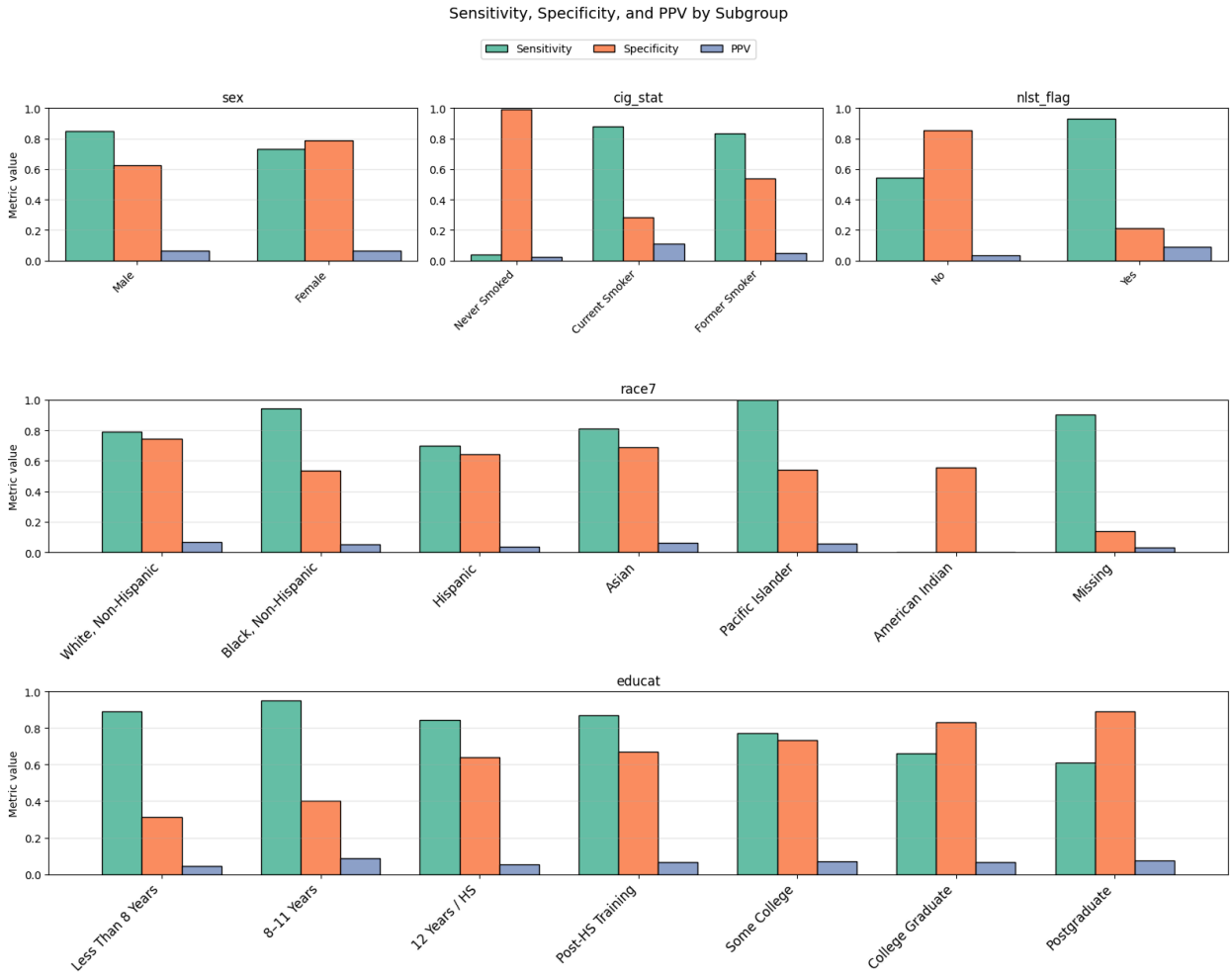
3. Metric-Specific Threshold: Choosing a threshold specifically to meet a predefined requirement for a single metric, such as achieving a minimum level of sensitivity.

We prioritized the sensitivity-based threshold, as our choice of classification threshold was guided by the clinical priority of minimizing false negatives. A missed cancer diagnosis poses a significant risk to patient health by delaying critical treatment. In contrast, a false positive, while leading to additional procedures like imaging or biopsies, is a more

manageable outcome. For this reason, we selected the threshold that achieves 80% sensitivity.

Q2.4.4

We evaluated model performance across demographic and clinical subgroups using sensitivity, specificity, and PPV. As expected, sensitivity was higher among current and former smokers, while never smokers showed almost no sensitivity but very high specificity because lung cancer is rare in this group. For race and ethnicity, results for smaller subgroups such as Pacific Islander and American Indian were less reliable, likely due to limited sample sizes.



Q2.5 Study Design Limitations

Our subgroup analyses reveal several limitations. First, the low prevalence of lung cancer in this dataset results in very low PPV across all groups, meaning most positive predictions would still need follow-up testing to confirm disease. Secondly the analysis relies on a single dataset that may differ from the data on which the model was trained. Differences in demographics, smoking behaviors, or disease prevalence can limit generalizability to other populations. Finally, results for smaller racial and ethnic groups, such as Pacific Islanders and American Indians, are based on very few cases, making their performance estimates unstable and difficult to interpret.

To address these limitations, future studies should focus on validating the model in larger and more diverse datasets, with deliberate inclusion of underrepresented groups to ensure stable subgroup estimates. It will also be important to report performance metrics with confidence intervals to clearly communicate the level of uncertainty, particularly for smaller subgroups. These steps are essential for building stronger evidence on how well the model performs across populations and for ensuring that screening guidelines are both robust and equitable.